

Rachael Lang

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Education

University of Wisconsin–Madison

Master of Science in Computer Science

May 2025

GPA: 3.6

University of Wisconsin–Madison

Bachelor of Science in Computer Science, Minor: Data Science

Dec 2023

GPA: 3.6

Technical Skills

- **Languages:** TypeScript, JavaScript, Java, Python, Kotlin, SQL, Bash
- **Front-End:** React, React Native, HTML, CSS, Material UI, Sass, Responsive & Accessible Design
- **Back-End & Tools:** Spring Boot, Node.js, REST APIs, Jest, React Testing Library, Cypress, Git, CI/CD

Work Experience

Target Corporation – Front-End Developer Intern

Jun 2024 – Aug 2024

- Designed and implemented a structured installation guide in TypeScript and React, building a task-driven checklist interface with dynamic state management to standardize hardware setup workflows across engineering teams
- Translated hardware configuration steps into an intuitive, formatted UI with grouped tasks, checkboxes, and clear visual hierarchy, reducing configuration friction by 20%
- Architected reusable UI components using Material UI, reducing duplicate UI code across new screens by 25% and improving design consistency
- Authored UI behavior tests using Jest and React Testing Library (render, screen queries), increasing front-end test coverage and strengthening regression protection during iterative development

Intel Corporation – Software Engineer Intern

Nov 2023 – Jan 2024

- Engineered a Python-based fingerprinting system to recursively scan 1000s of recipe files, generate MD5 hashes, and detect configuration changes across manufacturing systems
- Designed SQL-backed change-detection logic comparing current file states against historical records to identify new, modified, and deleted key-value pairs with automated flagging
- Deployed the tool as a scheduled job supporting 30+ engineers during new product introduction cycles, reducing manual debugging time by an estimated 30%
- Optimized scan performance by leveraging file modification timestamps to eliminate redundant hash computation across large directory hierarchies

Target Corporation – Full-Stack Developer Intern

Jun 2023 – Aug 2023

- Implemented backend authorization logic in Kotlin and Java within Spring Boot services to enforce transaction and period-limit rules across thousands of daily payment validations
- Built and integrated REST API endpoints enabling communication between point-of-sale systems and payment issuers, validating 100+ request/response flows during development
- Engineered structured input validation and error-handling mechanisms to prevent invalid transaction states and improve reliability of authorization workflows
- Developed SQL queries supporting transaction data retrieval and rule evaluation, improving query clarity and maintainability across backend services

The J.M. Smucker Company – Software Engineer Intern

May 2022 – Aug 2022

- Led database schema updates in Oracle SQL to support modernization of a supply planning application during enterprise system migration efforts
- Developed an automated Bash-based file transfer solution with logging, retry logic, and compression to stabilize nightly multi-GB forecast file transfers to an external SaaS planning system
- Reduced after-hours manual intervention by 50% by preventing recurring file transfer failures in automated supply planning data workflows

Projects

RachaelTLang.com – Full-Stack Developer

Dec 2023 – Present

- Engineered a high-performance personal web application using HTML, CSS, and JavaScript, achieving 96+ Lighthouse performance scores through asset optimization, structured DOM design, and efficient rendering practices
- Designed a reproducible Git-based deployment workflow using GitHub Pages, enabling automated production releases and streamlined iteration without manual hosting configuration